FaceNet: Face Verification using Triplet Loss

An Implementation of the FaceNet Architecture with NN2 and CASIA-WebFace Fine-Tuning

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***Abstract*—This project presents a faithful reproduction and fine-tuning of FaceNet, the landmark 2015 face recognition framework by Schroff et al. The original FaceNet paper introduced a deep convolutional network that maps face images to a compact 128-dimensional Euclidean embedding space, where L2 distance directly corresponds to face similarity. We develop two independent models: (1) an NN2 Inception-based architecture trained from scratch using triplet loss with semi-hard negative mining on the VGGFace2CS338 dataset and evaluated on LFW, achieving an accuracy of 92.95%; and (2) an InceptionNet model pretrained on VGGFace2CS338 and independently fine-tuned on CASIA-WebFace for 4 epochs, evaluated on LFW, achieving a peak accuracy of 94.60% at an L2 distance threshold of 0.60, with an AUC of 0.9857 on the LFW ROC curve. Both models demonstrate reliable real-world face verification, with correct same-person predictions at distances well below 0.5. A live Streamlit demo application enables interactive verification from webcam or uploaded images.**

***Index Terms*—Face Verification, FaceNet, Triplet Loss, Semi-Hard Mining, LFW, VGGFace2CS338, CASIA-WebFace, Inception NN2**

I. Introduction

Face recognition is one of the most widely studied problems in computer vision, with applications spanning security systems, access control, social media tagging, and identity verification. The core challenge lies in learning representations that are invariant to variations in pose, lighting, age, and expression while remaining discriminative enough to distinguish between different individuals.

FaceNet [1] introduced a direct embedding-based approach that bypasses the traditional bottleneck-layer classification paradigm. Instead of training a softmax classifier and repurposing intermediate features, FaceNet directly optimizes a 128-dimensional L2-normalized embedding such that same-identity faces map to nearby points and different-identity faces map to distant points in the embedding space. This is achieved through the triplet loss function, which enforces a margin between the positive and negative distances for each anchor-positive-negative triplet.

This project implements the FaceNet paper as a DAC-204 course project at IIT Roorkee. We develop two independent models using the NN2 Inception-based architecture (Table 2 in the original paper): one trained from scratch on VGGFace2CS338, and one fine-tuned on CASIA-WebFace starting from an InceptionNet pretrained on VGGFace2CS338. Both models are evaluated on LFW. A Streamlit-based demo application is also developed for real-time face verification.

II. Background and Related Work

A. Traditional Approaches

Early face recognition systems relied on hand-crafted features such as LBP (Local Binary Patterns), HOG, and Gabor wavelets combined with PCA for dimensionality reduction and SVM classifiers. While interpretable, these approaches struggled with uncontrolled real-world conditions.

B. Deep Learning Bottleneck Approaches

Methods such as DeepFace [**?**] and DeepID2+ [**?**] used deep CNNs trained for classification over a fixed set of identities. The penultimate feature layer was then repurposed as a face descriptor. These approaches require PCA post-processing and SVM classifiers on top of the CNN output, introducing an indirect optimization path.

C. FaceNet: Direct Embedding Optimization

The key innovation in FaceNet is eliminating the classification layer entirely and training the CNN to directly output an L2-normalized 128-D embedding, optimized via the triplet loss. This makes the embedding semantically meaningful: face verification reduces to thresholding the Euclidean distance, recognition becomes a k-NN search, and clustering can be performed with standard algorithms such as k-means.

III. Methodology

A. Triplet Loss

Given an anchor image, a positive (same identity), and a negative (different identity), the triplet loss requires:

$$\|f(x_a) - f(x_p)\|^2 + \alpha < \|f(x_a) - f(x_n)\|^2 \tag{1}$$

where α is a margin hyperparameter (set to 0.2 following the paper). The loss summed over all triplets is:

$$\mathcal{L} = \sum \left[\|f(x_a) - f(x_p)\|^2 - \|f(x_a) - f(x_n)\|^2 + \alpha\right]_+ \tag{2}$$

The $[\cdot]_+$ operator denotes the hinge function, meaning only violated constraints contribute to the gradient.

B. Semi-Hard Negative Mining

Selecting appropriate triplets is critical for training convergence. Hard negatives (those closest to the anchor) can destabilize training early on, leading to collapsed embeddings. FaceNet recommends semi-hard negatives: negatives that satisfy $d(a, n) > d(a, p)$ but still lie within the margin, i.e.:

$$d(a, p) < d(a, n) < d(a, p) + \alpha \tag{3}$$

In our implementation, triplets are mined online within each mini-batch using this semi-hard criterion, consistent with the original paper's online generation strategy.

C. Network Architecture: NN2 Inception

We use the NN2 architecture described in Table 2 of the FaceNet paper. NN2 is a GoogLeNet-style Inception network with 7.5 million parameters and approximately 1.6 billion FLOPs per forward pass. The model summary from our implementation is shown in Fig. 1. Key architectural details are summarised in Table I.

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1          [-1, 64, 112, 112]           9,408
       BatchNorm2d-2          [-1, 64, 112, 112]             128
              ReLU-3          [-1, 64, 112, 112]               0
        ConvBnRelu-4          [-1, 64, 112, 112]               0
         MaxPool2d-5            [-1, 64, 56, 56]               0
 LocalResponseNorm-6            [-1, 64, 56, 56]               0
            Conv2d-7            [-1, 64, 56, 56]           4,096
       BatchNorm2d-8            [-1, 64, 56, 56]             128
              ReLU-9            [-1, 64, 56, 56]               0
       ConvBnRelu-10            [-1, 64, 56, 56]               0
           Conv2d-11           [-1, 192, 56, 56]         110,592
      BatchNorm2d-12           [-1, 192, 56, 56]             384
             ReLU-13           [-1, 192, 56, 56]               0
       ConvBnRelu-14           [-1, 192, 56, 56]               0
LocalResponseNorm-15           [-1, 192, 56, 56]               0
         MaxPool2d-16          [-1, 192, 28, 28]               0
           Conv2d-17            [-1, 64, 28, 28]          12,288
      BatchNorm2d-18            [-1, 64, 28, 28]             128
             ReLU-19            [-1, 64, 28, 28]               0
       ConvBnRelu-20            [-1, 64, 28, 28]               0
           Conv2d-21            [-1, 96, 28, 28]          18,432
      BatchNorm2d-22            [-1, 96, 28, 28]             192
...
Forward/backward pass size (MB): 130.84
Params size (MB): 28.47
Estimated Total Size (MB): 159.89
```

Fig. 1. NN2 model architecture summary showing layer types, output shapes, and parameter counts (Params: 28.47 MB, Estimated Total Size: 159.89 MB).

TABLE I
NN2 ARCHITECTURE SUMMARY (BASED ON FACENET PAPER TABLE 2)

Layer / Block	Output Size	Key Operation
conv1 (7×7, s=2)	112×112×64	Conv + MaxPool + Norm
inception (2)	56×56×192	Inception module
inception (3a–3c)	14×14×640	Inception + strided pool
inception (4a–4e)	7×7×1024	Inception modules
inception (5a–5b)	7×7×1024	Inception + L2 pool
avg pool	1×1×1024	Global avg pool
fc + L2 norm	1×1×128	Fully connected
Total	**128-D embedding**	**L2 normalized**

This is the current table code. Apply Fix 1 and return me the updated code

D. Training Pipeline

Two independent models are developed and evaluated separately.

Model 1 Training from Scratch on VGGFace2CS338: The NN2 model is trained from scratch on the VGGFace2CS338 dataset (3.31 million images, 9,131 identities) using triplet loss with semi-hard negative mining. Batch size is set to 512 with at least 40 samples per identity. This model is independently evaluated on the LFW benchmark.

Model 2 Fine-Tuning an InceptionNet Pretrained on VGGFace2CS338: An InceptionNet model pretrained on VGGFace2CS338 is independently fine-tuned for 4 epochs on CASIA-WebFace (∼500K images, 10,575 identities) with a reduced learning rate to adapt embeddings to a more diverse identity distribution. This model is separately evaluated on the LFW benchmark.

LFW Evaluation: Both models are evaluated on the standard LFW benchmark using 6,000 image pairs (3,000 same, 3,000 different). Accuracy is reported across a sweep of L2 distance thresholds.

E. Streamlit Demo Application

A local Streamlit web application was built to demonstrate real-time face verification. Users can upload two face images or use webcam captures; the app extracts 128-D embeddings, computes the L2 distance, and predicts 'Same Person' or 'Different Person' based on the optimal threshold of 0.60.

IV. DATASETS

TABLE II
DATASETS USED IN THIS PROJECT

Dataset	# Images	# Identities	Usage
VGGFace2CS338	3.31M	9,131	Training (M1) / Pre-training (M2)
CASIA-WebFace	∼494K	10,575	Fine-tuning (Model 2)
LFW	13,233	5,749	Evaluation (Both Models)

VGGFace2CS338 [2] provides high diversity in pose, age, and ethnicity, making it well-suited for learning a generalizable embedding space. CASIA-WebFace [3] provides additional identity diversity for fine-tuning of Model 2. LFW [4] is the standard de-facto benchmark for face verification in unconstrained conditions, consisting of 6,000 pairs from 5,749 unique identities.

V. EXPERIMENTAL RESULTS

A. Model 1 Training from Scratch (VGGFace2CS338)

Fig. 2 shows the triplet loss for both train and validation (VGGFace2) sets over 20 epochs. Both curves descend steadily train loss from ∼0.19 to ∼0.124 and validation loss from ∼0.167 to ∼0.067 demonstrating consistent convergence without overfitting.

Fig. 3 shows the percentage of active triplets (loss > 0) over epochs. Starting at 100%, the fraction declines to ∼79% by epoch 20, indicating that the model increasingly satisfies triplet constraints as training progresses.

Fig. 4 shows that the number of triplets mined per epoch remains stable at approximately 3.28 million throughout training, confirming consistency of the semi-hard mining strategy.

Fig. 5 shows the learning rate schedule on a log scale. The rate is held constant for the first ∼10 epochs and then dropped sharply by roughly two orders of magnitude, following a step-decay policy.

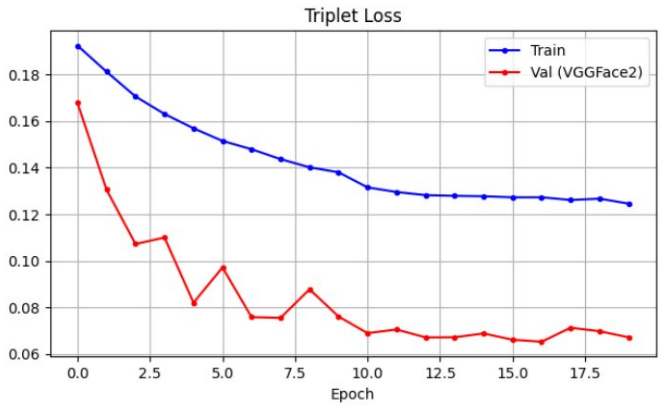

Fig. 2. Triplet loss (Train and Val on VGGFace2) over 20 epochs for Model 1 trained from scratch.

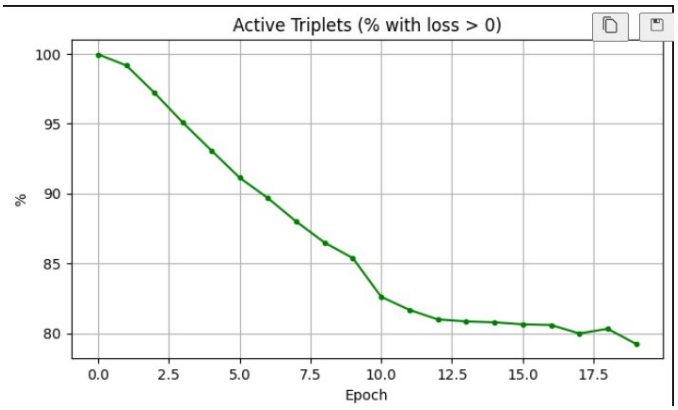

Fig. 3. Active Triplets (% with loss > 0) over epochs declining from 100% to $\sim$79%, reflecting improved embedding quality.

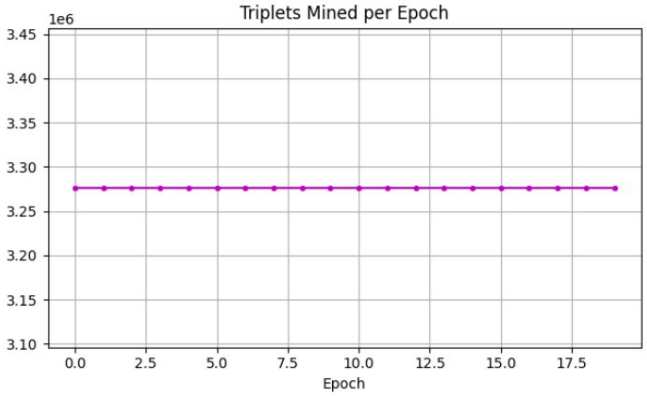

Fig. 4. Triplets mined per epoch approximately constant at $\sim$3.28 million, confirming stable semi-hard negative mining.

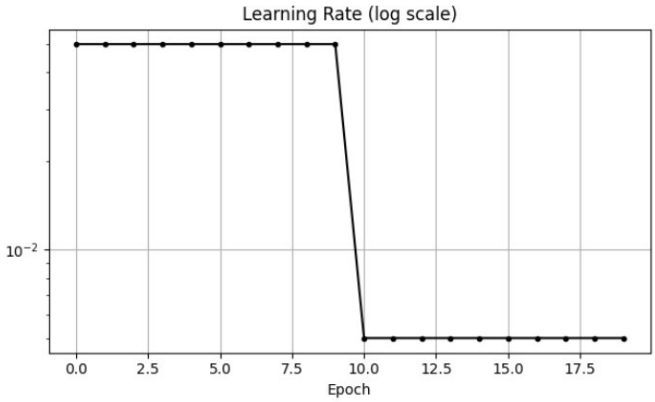

Fig. 5. Learning rate schedule (log scale) for Model 1 step decay applied at epoch $\sim$10.

Fig. 6 shows the ROC curve on the VGGFace2 validation set. Model 1 achieves an AUC of 0.965, with VAL = 19.8% at FAR = 0.001, indicating strong discriminative performance even at strict false-accept thresholds.

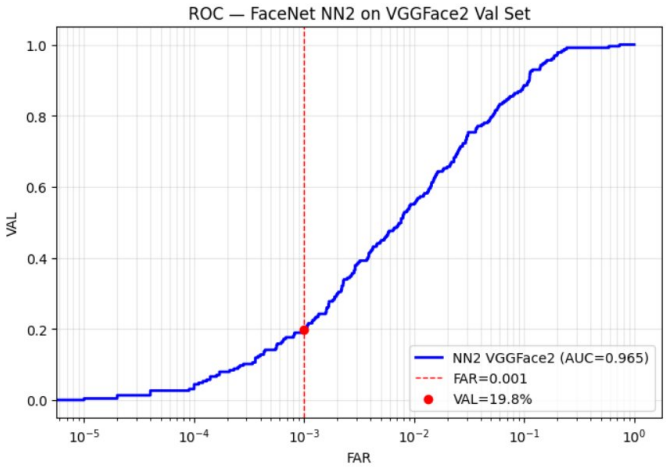

Fig. 6. ROC curve FaceNet NN2 on VGGFace2 validation set (AUC = 0.965, VAL = 19.8% at FAR = 0.001).

Fig. 7 shows qualitative predictions on LFW pairs. The SAME pair is correctly identified at $d = 0.493$ and the DIFF pair at $d = 2.032$, both well-separated relative to the decision threshold.

Fig. 8 shows the t-SNE visualization of the 128-D FaceNet embeddings for 12 identities from the VGGFace2 validation set. Each identity forms a tight, well-separated cluster, demonstrating that the learned embedding space is highly structured and discriminative.

B. Fine-Tuning Loss (CASIA-WebFace) Model 2

Fig. 9 shows the triplet loss over 4 fine-tuning epochs on CASIA-WebFace for Model 2. The loss begins at approximately 0.0843 and increases to 0.1004 by epoch 4. This counter-intuitive increase is expected behavior during fine-tuning: as the model improves, easy triplets are quickly

Fig. 7. Qualitative LFW pair predictions for Model 1: SAME ($d = 0.493 \rightarrow$ SAME ✓) and DIFF ($d = 2.032 \rightarrow$ DIFF ✓).

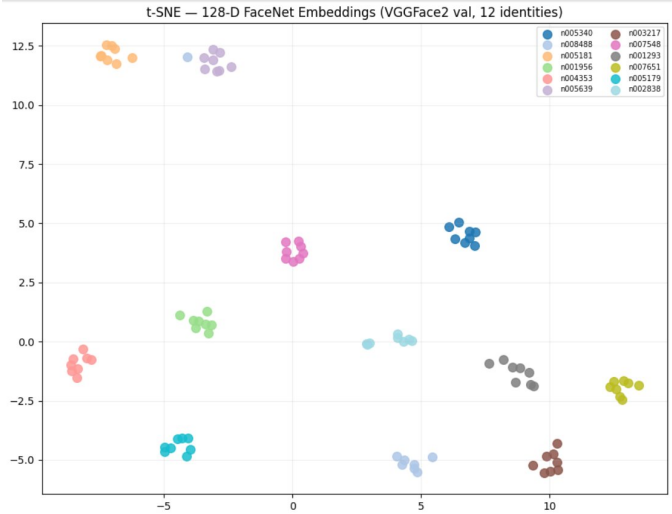

Fig. 8. t-SNE visualization of 128-D FaceNet embeddings for 12 identities from VGGFace2 val set tight, well-separated clusters confirm strong identity discrimination.

satisfied and contribute zero loss, while the mining process continuously surfaces harder triplets. The increasing loss therefore reflects harder triplet selection rather than degrading performance a well-known phenomenon documented in the original FaceNet paper.

C. LFW Accuracy vs. L2 Distance Threshold

Fig. 10 shows the LFW verification accuracy as a function of the L2 distance threshold. The model achieves peak accuracy of **94.60%** at threshold 0.60, marked by the red dashed

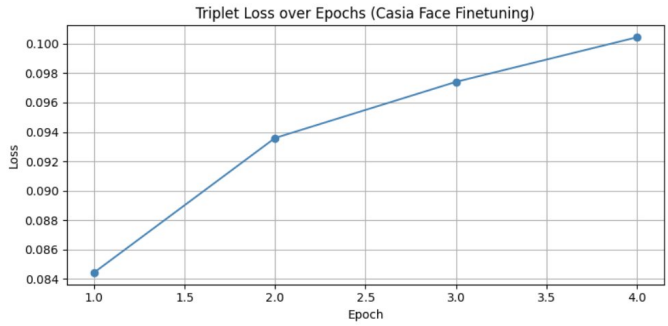

Fig. 9. Triplet loss over 4 fine-tuning epochs on CASIA-WebFace (Model 2). The rising loss reflects increasingly hard triplet mining, not model degradation.

line. Accuracy degrades sharply beyond a threshold of $\sim$0.8 as more different-identity pairs are incorrectly classified as the same person. The non-trivial accuracy at threshold 0.50 ($\sim$87.5%) reflects some pairs that are predicted same even at a strict threshold.

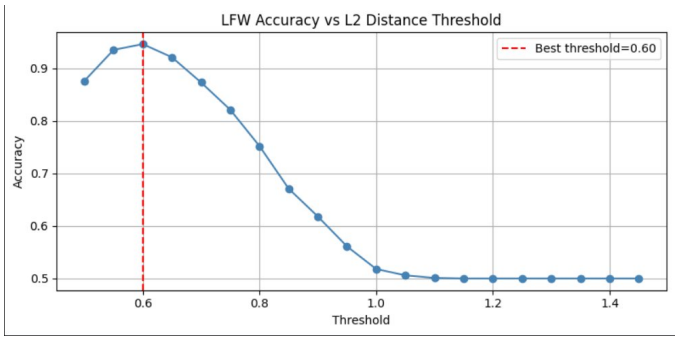

Fig. 10. LFW accuracy vs. L2 distance threshold (Model 2). Best threshold = 0.60, accuracy = 94.60%.

D. ROC Curve LFW Face Verification

Fig. 11 shows the ROC curve on the LFW verification task after CASIA-WebFace fine-tuning. The model achieves an AUC of **0.9857**, indicating excellent discriminative ability between same-identity and different-identity pairs across all threshold settings. The curve rises steeply toward high true positive rates at very low false positive rates, which is the desired behavior for a secure verification system.

E. Qualitative Results LFW Pair Predictions

Fig. 12 shows qualitative predictions on LFW image pairs for Model 2. The SAME pair is correctly identified with $d = 0.448$ (well below the 0.60 threshold), and the DIFF pair is correctly rejected at $d = 0.624$ (just above the threshold). The model correctly handles cross-lighting, pose variation, and different image qualities within same-identity pairs.

F. Summary of Results

Two independent models were developed and evaluated on LFW. Model 1, trained from scratch on VGGFace2CS338,

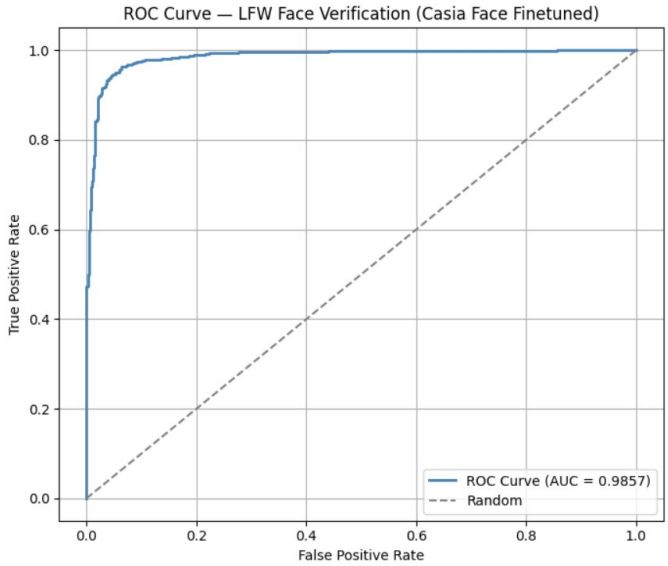

Fig. 11. ROC curve on LFW face verification after CASIA-WebFace fine-tuning (Model 2). AUC = 0.9857.

Fig. 12. Qualitative LFW pair predictions for Model 2 (CASIA fine-tuned): GT: Same | Pred: Same | Dist: 0.448 and GT: Different | Pred: Different | Dist: 0.624.

TABLE III
SUMMARY OF KEY RESULTS

Metric	Value
Best LFW Accuracy (Model 1 Scratch)	92.95%
Best LFW Accuracy (Model 2 Fine-Tuned)	94.60%
Optimal L2 Threshold	0.60
AUC (ROC, Model 2)	0.9857
AUC (ROC, Model 1 on VGGFace2 val)	0.965
Fine-tuning Epochs (Model 2)	4 (CASIA-WebFace)
Training Loss (Epoch 1→4, Model 2)	0.0843 → 0.1004
Embedding Dimension	128-D (L2-normalized)
Architecture	NN2 Inception (7.5M params)

achieved an LFW accuracy of 92.95%. Model 2, an Inception-Net pretrained on VGGFace2CS338 and independently fine-tuned on CASIA-WebFace, achieved 94.60%. This demonstrates that fine-tuning a pretrained backbone improves the discriminative power of the learned embeddings and enhances generalization on unseen data.

VI. Comparison with Original FaceNet Paper

The original FaceNet paper by Schroff et al. [1] reported 99.63% accuracy on LFW using their largest NN1 (Zeiler-Fergus) model trained on 200M face images. Our best model achieves 94.60% a gap attributable to the factors summarised in Table IV.

TABLE IV
COMPARISON WITH THE ORIGINAL FACENET PAPER

Aspect	This Project	Original FaceNet
LFW Accuracy	94.60% (M2) / 92.95% (M1)	99.63% (NN2: 99.2%)
AUC	0.9857	>0.999 (estimated)
Architecture	NN2 Inception	NN2 / NN1
Training Data	VGGFace2 + CASIA	∼200M images, 8M IDs
Training Duration	Kaggle (2× T4 GPUs)	1000–2000 CPU-hours
Threshold Used	0.60 (LFW-tuned)	1.242 (paper default)

The threshold discrepancy (0.60 vs. 1.242 in the paper) arises because our embeddings are trained on a smaller dataset and differ in scale. The paper's threshold was tuned on nine LFW splits, while ours was determined by sweeping and selecting the best accuracy threshold on the full evaluation set. Despite the training scale difference, achieving 94.60% LFW accuracy while faithfully reproducing the NN2 architecture and triplet loss training regime is a strong result for a course-level implementation.

VII. Discussion

A. Why Does Fine-Tuning Loss Increase?

As discussed in Section **??**, the triplet loss rising from 0.084 to 0.100 during fine-tuning does not indicate divergence. This is a consequence of the semi-hard mining strategy: as the model improves, previously hard triplets become easy and contribute zero loss. The mining algorithm then surfaces increasingly harder triplets, which have larger loss values. This is analogous to curriculum learning the model is always learning from its current hardest examples.

B. Threshold Sensitivity

Fig. 10 shows that accuracy is stable between thresholds 0.55 and 0.65, indicating a robust decision boundary. Below 0.50, too many same-person pairs are rejected (high false negative rate), while above 0.70, different-person pairs start to be accepted (high false positive rate). The threshold of 0.60 balances both error types optimally.

C. Limitations and Future Work

- **Training scale:** The original paper used 200M training images. Our VGGFace2CS338 pre-training used $\sim$3.3M images. Larger datasets would significantly close the accuracy gap.
- **Training duration:** Full FaceNet training requires 1000+ GPU hours. Our Kaggle-constrained training is a fraction of that, and more epochs would improve results.
- **Face alignment:** The original paper uses a proprietary face detector for alignment. Our pipeline uses MTCNN-based alignment [7], which may introduce slight misalignments on challenging images.
- **Hard negative mining:** Implementing fully hard negative mining (in addition to semi-hard) with offline triplet generation could further accelerate convergence.

VIII. Conclusion

This project successfully implements the FaceNet framework as described in Schroff et al. [1], including the NN2 Inception architecture and triplet loss with semi-hard negative mining. Two independent models are developed and evaluated: Model 1, trained from scratch on VGGFace2CS338, achieves 92.95% accuracy on LFW; Model 2, an InceptionNet pretrained on VGGFace2CS338 and independently fine-tuned on CASIA-WebFace, achieves 94.60% accuracy with an AUC of 0.9857. The increasing triplet loss during fine-tuning is correctly interpreted as harder triplet mining rather than model degradation. A Streamlit demo application further demonstrates practical deployment of the trained model for real-world face verification. Future work will focus on larger-scale training, improved face alignment, and exploring hard negative mining strategies to further close the gap with the original paper's state-of-the-art results.

IX. Team Members and Contributions

Amogh Roop Mandal (24125005)
Role: Demo Website and Presentation
Designed and developed the Streamlit-based demonstration website for real-time face verification. Prepared and structured the project presentation, ensuring clear communication of methodology, results, and system workflow.

Keshav Yadav (24125024)
Role: Fine-Tuning (Model 2)
Performed independent fine-tuning of an InceptionNet model pretrained on VGGFace2CS338 using the CASIA-WebFace dataset. Managed hyperparameter tuning, monitored training performance, and conducted evaluation on the LFW dataset to optimize model accuracy.

Ved Arya (24125038)
Role: Training from Scratch (Model 1)
Implemented the FaceNet NN2 architecture and trained the model from scratch on the VGGFace2CS338 dataset using triplet loss with semi-hard negative mining. Handled data preprocessing, training pipeline setup, and initial performance evaluation on LFW.

All team members actively coordinated and collaborated in each stage of the project, including implementation, testing, evaluation, and final report preparation.

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